

CLASS 12 · CHEMISTRY · SCHOOL TEST PAPER

ChemPaper

Worked Solutions

All four sections answered, with the reasoning shown

Sections A–D · 62 marks

Every alternative worked

Numericals recomputed

What this is. The paper in `Chemistry/ChemPaper.pdf`, read off the scan and answered question by question. Each answer opens with *How to see it* — the one idea being tested — then the working in the order an examiner expects to read it, then the answer. Where the paper offers an internal choice, **both** alternatives are worked, so nothing here depends on which one you picked.

The paper is a scan. It carries no text layer, so it was read from the page images and checked against OCR. The corrected transcript is `Chemistry/OCR/chempaper.txt`. Four places where the printed paper is wrong or incomplete are named on the next page rather than quietly fixed.

How to use it. Sit the paper first, then mark yourself against this. In a numerical the method carries most of the marks: write the formula, then the formula with the numbers in it, then the answer with its unit.

Before you start

Four things the printed paper gets wrong or leaves out. They are named here so that an answer below never looks like it is quietly disagreeing with the question.

The assertion–reason options are missing

Items x to xiv are assertion–reason questions, six marks in all, but the four options they are to be answered from are never printed. They are supplied on the next page and used throughout.

Q15, experiment 2: the rate is mistyped

The table prints 7×10^{-2} . It should be 7.2×10^{-2} . With 7.2 all four experiments give $k = 6.00$ exactly; with 7.0 experiment 2 alone gives 5.83, and no rate law fits all four rows. Q15 is solved with 7.2.

Q12(a) has no instruction

It prints only the statement — that $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3$ with HI gives CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$ — and stops. It is the standard *explain why*, and is answered as such.

Section B Q1 is missing a word

“the acid catalysed *of* ethanol to ethene” — the word is **dehydration**. Also: LiAlH_4 is LiAlH_4 , KMO_4 is KMnO_4 , Cl_2 is Cl_2 , $\text{S}_{\text{N}}\text{I}$ is $\text{S}_{\text{N}}\text{I}$ and “emi” is emf.

How the 62 marks divide

Section	Questions	Each	Marks
A	Q1 (i)–(xv)	1	15
B	Q1–Q7	2	14
C	Q8–Q13	3	18
D	Q14–Q16	5	15
		Total	62

Section B restarts its numbering at 1; Section C then continues from 8. The paper prints no total, no time allowed and no general instructions.

Section A · 15 marks

Question 1 — fifteen sub-parts, one mark each. Nine multiple choice, six assertion–reason (one with an internal choice).

Multiple choice

9 × 1 = 9

1 (i) Haloarenes

[1]

Conversion of chlorobenzene into phenol involves:

- (a) modified S_N1 mechanism
- (b) modified S_N2 mechanism
- (c) both (a) and (b)

(d) elimination–addition mechanism

SOLUTION

How to see it · Ask what chlorobenzene *cannot* do first. S_N1 needs an aryl cation, which would put a positive charge in an sp² orbital in the ring plane — not stabilised by the π system at all. S_N2 needs backside attack, and the benzene ring is sitting exactly where the nucleophile would have to come from. So neither of the ordinary routes is open, and the reaction must go by some third path.

1. The C–Cl bond in chlorobenzene has **partial double-bond character** (the chlorine lone pair is delocalised into the ring), so it is shorter and stronger than in an alkyl halide.
2. Under Dow's conditions — NaOH, **623 K, 300 atm** — the hydroxide is basic enough to pull off an *ortho* hydrogen. Cl[–] then leaves, and what is left is **benzyne**, a ring carrying an extra, very strained π bond.
3. OH[–] adds across that triple bond, and protonation gives phenoxide, then phenol on acidification.
4. Base removes H, halide leaves → that is an **elimination**; nucleophile then adds to benzyne → an **addition**. Hence elimination–addition.

Answer · (d) elimination–addition, i.e. the benzyne mechanism.

The evidence is a labelling experiment: chlorobenzene labelled with ¹⁴C at C-1 gives phenol with the label split 50:50 between C-1 and C-2. Only a symmetrical intermediate — benzyne — can do that.

Common slip · If the ring carries strong electron-withdrawing groups *ortho* or *para* to the halogen (2,4-dinitrochlorobenzene, say), the mechanism switches to **addition–elimination** through a Meisenheimer complex, and the conditions become mild. The question says plain chlorobenzene, so it is the benzyne route.

Phenol on treatment with excess of conc. HNO_3 gives:

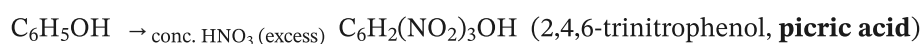
- (a) o-nitrophenol
- (b) p-nitrophenol
- (c) o- and p-nitrophenol

(d) 2,4,6-trinitrophenol

SOLUTION

How to see it · The word doing the work is **excess**. $-\text{OH}$ is a powerful activating, *o,p*-directing group, so once the first nitro group is in, the ring is still reactive enough to take more. Dilute HNO_3 stops at mononitration; concentrated acid in excess goes all the way.

1. Phenol + dilute HNO_3 , 298 K $\rightarrow$ a mixture of *o*- and *p*-nitrophenol (that is option (c), and it is the answer to a *different* question).
2. Phenol + **conc. HNO_3 in excess** $\rightarrow$ substitution at both *ortho* positions and the *para* position.



Answer · **(d)** 2,4,6-trinitrophenol (picric acid).

$\text{CH}_3\text{CH}_2\text{OH}$ can be converted into CH_3CHO by:

- (a) catalytic hydrogenation
- (b) treatment with LiAlH_4

(c) treatment with PCC in CH_2Cl_2

- (d) treatment with KMnO_4/H^+

SOLUTION

How to see it · Alcohol $\rightarrow$ aldehyde is an **oxidation that must be stopped half-way**. Two of the options reduce instead of oxidising, and one oxidises too far. Only one reagent is mild enough to stop at the aldehyde.

1. **(a)** and **(b)** are *reductions* — they run the wrong way. LiAlH_4 would turn an aldehyde back into the alcohol.
2. **(d)** acidified KMnO_4 is a strong oxidant: it does not stop at CH_3CHO but carries straight on to CH_3COOH .
3. **(c)** pyridinium chlorochromate in dichloromethane is the standard mild oxidant. It has no water in it, so the aldehyde is never hydrated and cannot be oxidised further.



Answer · **(c)** treatment with PCC in CH_2Cl_2 .

Which mixture shows positive deviation from Raoult's law?

- (a) Acetone + Aniline
(b) **Acetone + Ethanol**
(c) Water + Nitric acid
(d) Chloroform + Benzene

SOLUTION

How to see it · One test settles all four. Positive deviation means the **A–B attraction is weaker** than the A–A and B–B attractions the components had on their own: molecules escape more easily, so the vapour pressure is higher than Raoult's law predicts. So look for a mixture in which mixing **breaks** hydrogen bonds and puts nothing as strong back.

1. **Acetone + ethanol.** Pure ethanol is extensively hydrogen bonded. Acetone gets between the ethanol molecules and breaks those bonds; acetone has no O–H of its own, so the new acetone–ethanol interaction is weaker than the ethanol–ethanol one it replaced. Vapour pressure rises → **positive deviation**.
2. **Acetone + aniline:** aniline's N–H hydrogen bonds to acetone's carbonyl oxygen — a *new* attraction is formed → negative.
3. **Water + nitric acid:** strong hydrogen bonding between the two → negative (it forms a maximum-boiling azeotrope).
4. **Chloroform + benzene:** the polarised C–H of chloroform interacts with the benzene π cloud → negative.

Answer · (b) Acetone + Ethanol.

Common slip · A positive deviation goes with a **minimum**-boiling azeotrope and $\Delta_{\text{mix}}H > 0$ (mixing absorbs heat); a negative deviation with a **maximum**-boiling azeotrope and $\Delta_{\text{mix}}H < 0$. Getting these the wrong way round is the commonest error on this topic.

Charge required to liberate 1 mol of $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ is:

- (a) 1 F
(b) 3 F
(c) **5 F**
(d) 4 F

SOLUTION

How to see it · Faraday's first law in one line: **one mole of electrons is one faraday**. So the whole question is “how many electrons does this half-reaction take?” — which is just the change in oxidation number of manganese.

1. In MnO_4^- : $4 \times (-2)$ for the oxygens = -8 , and the ion is -1 , so Mn is **+7**.
2. In Mn^{2+} manganese is **+2**.
3. Change = $7 - 2 = 5$, so 5 mol of electrons per mole of permanganate.



Answer · (c) 5 F ($= 5 \times 96500 = 482500$ C).

Henry's law constant is larger when gas solubility is:

- (a) Higher
- (b) Lower**
- (c) Independent
- (d) Pressure-dependent

SOLUTION

How to see it · Write the law down and read it. $p = K_H x$, so $K_H = p/x$: the constant is **inversely** related to the mole fraction dissolved. Large K_H means you need a large pressure to dissolve very little — a poorly soluble gas.

$$p = K_H \cdot x \Rightarrow K_H = \frac{p}{x} \text{ so } K_H \uparrow \Leftrightarrow x \downarrow$$

Answer · (b) Lower.

This is why K_H increases with temperature — gases get less soluble as you warm the liquid. Item (xi) below is the same fact asked as an assertion–reason.

Deficiency of which vitamin caused rickets?

- (a) Vitamin C
- (b) Vitamin B
- (c) Vitamin A
- (d) Vitamin D**

SOLUTION

How to see it · Rickets is soft, deformed **bone**. Bone is calcium phosphate, and the vitamin that governs calcium absorption from the gut is D.

- **D** → rickets (children), osteomalacia (adults)
- **C** → scurvy — bleeding gums
- **A** → xerophthalmia, night blindness
- **B₁** → beri-beri

Answer · (d) Vitamin D.

The values of van't Hoff factor for KCl, K₂SO₄ and C₆H₁₂O₆ is

- (a) 2, 2 and 2
- (b) 2, 3 and 2
- (c) 3, 2 and 1

(d) 2, 3 and 1

SOLUTION

How to see it · i is just **how many particles one formula unit gives** in solution, assuming complete dissociation. Count the ions; a non-electrolyte counts as one.

1. $\text{KCl} \rightarrow \text{K}^+ + \text{Cl}^- \rightarrow 2$ particles.
2. $\text{K}_2\text{SO}_4 \rightarrow 2\text{K}^+ + \text{SO}_4^{2-} \rightarrow 3$ particles.
3. C₆H₁₂O₆ (glucose) is a non-electrolyte — it dissolves as whole molecules $\rightarrow 1$.

Answer · (d) 2, 3 and 1.

Common slip · $i > 1$ means dissociation, $i < 1$ means **association** — benzoic acid dimerising in benzene, as in Q11 of Section C, has $i \approx 0.5$.

The reaction of toluene with Cl₂ in presence of FeCl₃ gives 'X' while the reaction of toluene with Cl₂ in presence of light gives 'Y'. Thus 'X' and 'Y' are

- (a) X = benzyl chloride, Y = o- and p-chlorotoluene
- (b) X = m-chlorotoluene, Y = p-chlorotoluene

(c) X = o- and p-chlorotoluene, Y = trichloromethylbenzene

- (d) X = benzyl chloride, Y = m-chlorotoluene

SOLUTION

How to see it · Two different conditions, two completely different mechanisms. **A Lewis acid attacks the ring; light attacks the side chain.** FeCl₃ polarises Cl₂ into an electrophile, so you get electrophilic substitution on the ring. Light homolyses Cl₂ into radicals, and a radical goes for the weakest C–H, which is the benzylic one.

1. **X, ring chlorination.** –CH₃ is activating and *o,p*-directing, so the product is a mixture of *o*- and *p*-chlorotoluene. *Not meta* — which rules out (b) and (d) at once.
2. **Y, side-chain chlorination.** The benzyl radical is stabilised by the ring, so substitution happens there. With excess Cl₂ it does not stop at benzyl chloride: it goes on through benzal chloride to **benzotrichloride** (trichloromethylbenzene), C₆H₅CCl₃.

Answer · (c) X = *o*- and *p*-chlorotoluene, Y = trichloromethylbenzene.

Common slip · Option (a) is the same two products with the labels **swapped**. The mnemonic is worth keeping: *light* goes for the *side chain*, a *Lewis acid* goes for the *ring*.

Assertion and reason

6 × 1 = 6

The assertion–reason options. The paper prints six assertion–reason items (x–xiv, one of them with an OR) but **never prints the option list**. These are the four standard options, and they are what the letters below refer to.

- a. Both A and R are true, and R **is** the correct explanation of A.

- b. Both A and R are true, but R is **not** the correct explanation of A.
c. A is true but R is false.
d. A is false but R is true.

1 (x) Phenols

[1]

Assertion (A): p-nitrophenol is more acidic than phenol.

Reason (R): $-\text{NO}_2$ group helps in the stabilisation of the phenoxide ion by dispersal of negative charge due to resonance.

SOLUTION

How to see it · Acidity of a phenol is decided by **how stable the phenoxide is**, not by the phenol. Anything that spreads the negative charge out makes the anion more stable and the acid stronger.

1. A is **true**: p-nitrophenol ($\text{pK}_a \approx 7.1$) is considerably more acidic than phenol ($\text{pK}_a \approx 10$).
2. R is **true**: $-\text{NO}_2$ is both $-I$ and $-R$. From the *para* position its resonance reaches the oxygen, so the negative charge of the phenoxide is delocalised onto the nitro oxygens as well as around the ring.
3. And R **is** the reason for A — that extra delocalisation is exactly why the anion is more stable and the acid stronger.

Answer · (a) — both true, R is the correct explanation of A.

Resonance from $-\text{NO}_2$ only reaches the ortho and para positions. m-nitrophenol is acidified by the $-I$ effect alone, so it sits between phenol and p-nitrophenol.

1 (xi) Solutions

[1]

Assertion (A): Henry's law constant increases with temperature.

Reason (R): Solubility of gas in liquid decreases with increase in temperature.

SOLUTION

How to see it · Same relation as item (vi): $K_H = p/x$. If warming the liquid drives gas out (smaller x at the same p), then K_H must get bigger. The two statements are not merely both true — they are the same fact stated two ways.

1. A is **true**: K_H rises with temperature.
2. R is **true**: dissolving a gas is exothermic, so by Le Chatelier's principle warming shifts the equilibrium back towards undissolved gas. It is why fizzy drinks go flat when warm and why aquatic life is stressed by thermal pollution.
3. R explains A directly, through $K_H = p/x$.

Answer · (a) — both true, R is the correct explanation of A.

Assertion (A): Isotonic solutions have same osmotic pressure.

Reason (R): 0.1 M NaCl and 0.1 M glucose will be isotonic.

SOLUTION

How to see it · A is the *definition* of isotonic, so it cannot be false. The whole question is whether R's example is a real one — and that turns on the van't Hoff factor, which is item (viii) again.

1. A is **true** — by definition, isotonic solutions are ones of equal osmotic pressure.
2. R is **false**. $\pi = iCRT$. Glucose is a non-electrolyte, $i = 1$, so $\pi \propto 0.1$. NaCl dissociates, $i = 2$, so $\pi \propto 0.2$ — **twice** as large. They are not isotonic; the NaCl solution is hypertonic to the glucose one.

Answer · (c) — A is true but R is false.

To make them isotonic you would need 0.05 M NaCl against 0.1 M glucose.

Assertion (A): Uracil is found in RNA.

Reason (R): RNA does not undergo replication.

SOLUTION

How to see it · Check the two statements separately, then ask the harder question: does the second one *cause* the first? Here both are true, but they are facts about different things — one about which base is present, one about what the molecule does.

1. A is **true**: RNA uses adenine, guanine, cytosine and **uracil**; DNA uses thymine (5-methyluracil) in place of uracil.
2. R is **true** in the sense meant here: DNA replicates, whereas cellular RNA is transcribed from DNA and is not itself replicated.
3. But R does **not** explain A. Uracil is present because RNA is synthesised from uridine nucleotides; the reason DNA uses thymine instead is a matter of *error correction* — cytosine deaminates to uracil, so if uracil were a legitimate DNA base that damage could not be recognised and repaired. Not being replicated is a consequence of RNA's role, not the cause of its base composition.

Answer · (b) — both true, but R is not the correct explanation of A.

Assertion (A): Rate and rate constant of zero order reactions have same units.

Reason (R): Half-life time of a zero-order reaction is independent of the initial concentration of reactant.

SOLUTION

How to see it · Two separate zero-order facts. Get the units from the rate law and the half-life from the integrated rate law — do not try to reason either of them out in words.

1. **A is true.** For zero order, rate = $k[A]^0 = k$. If rate is in $\text{mol L}^{-1} \text{s}^{-1}$ then so is k . They are numerically and dimensionally the same.
2. **R is false.** The integrated law is $[A] = [A]_0 - kt$. Putting $[A] = [A]_0/2$ gives
$$t_{1/2} = \frac{[A]_0}{2k} \quad \text{— directly proportional to } [A]_0$$
3. So the half-life of a zero-order reaction **does** depend on the initial concentration. It is the **first**-order half-life, $t_{1/2} = 0.693/k$, that is independent of it.

Answer · (c) — A is true but R is false.

Common slip · The statement in R is the one true, famous fact about first-order half-lives, transplanted onto zero order. Reading it too quickly is exactly what this item is testing.

Assertion (A): Reactions happen faster at higher temperatures.

Reason (R): As temperatures increase, there is also an increase in the number of collisions with the required activation energy for a reaction to occur.

SOLUTION

How to see it · The temptation is to say “molecules move faster, so they collide more often”. That is true but it is **not** the reason — a 10 K rise increases the collision frequency by only about 2%, yet it roughly doubles the rate. What actually changes is the *fraction* of collisions that are energetic enough.

1. A is **true**.
2. R is **true**, and it is carefully worded: not more collisions, but more collisions *with the required activation energy*.
3. On the Maxwell-Boltzmann curve, raising T flattens and widens the distribution, so the area beyond E_a — the fraction of molecules that can react — grows sharply. That is the $e^{-E_a/RT}$ term in the Arrhenius equation.
4. So R is the correct explanation of A.

Answer · (a) — both true, R is the correct explanation of A.

1 (xv) Chemical kinetics

[1]

For the reaction $A \rightarrow B$, the rate of reaction is quadrupled when the concentration of A is doubled; the rate of the reaction is $r = [A]^n$. The value of n is:

- (a) 1
- (b) 0
- (c) 3

(d) 2**SOLUTION**

How to see it · Order is the **power** the concentration is raised to, so take the ratio of the two rate expressions and everything but the factor cancels.

1. $r_1 = k[A]^n$ and $r_2 = k(2[A])^n$.
2. Divide: $r_2/r_1 = 2^n$. We are told this ratio is 4.
3. $2^n = 4 = 2^2 \Rightarrow n = 2$.

Answer · **(d)** $n = 2$ — the reaction is second order in A.

The same trick answers Q15(b) in Section D, where the concentration is tripled in a second-order reaction: $3^2 = 9$.

Section B · 14 marks

Seven questions of two marks. Four of them carry an internal choice; both alternatives are worked.

Question 1 · Dehydration of ethanol

2

B1 Alcohols

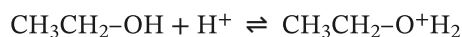
[2]

Write down the mechanism for the acid catalysed **dehydration** of ethanol to ethene.

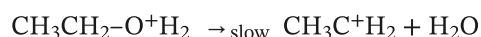
SOLUTION

How to see it · Three steps, and they are always the same three for an acid-catalysed dehydration: **protonate, lose water, lose a β -hydrogen**. The point of protonating first is that $-\text{OH}$ is a terrible leaving group, whereas $-\text{OH}_2^+$ leaves as neutral water, which is an excellent one.

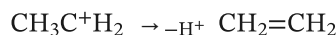
1. **Step 1 — protonation** (fast, reversible). The lone pair on the alcohol oxygen picks up H^+ from the acid, giving an ethyl oxonium ion.



2. **Step 2 — loss of water** (slow, **rate determining**). The C–O bond breaks heterolytically and water leaves, giving a carbocation.



3. **Step 3 — loss of a β -hydrogen** (fast). A base in the mixture — HSO_4^- or another water molecule — removes an H from the carbon *next* to the positive one. That pair of electrons becomes the π bond, and the catalyst is returned.



Answer · Overall, with conc. H_2SO_4 at **443 K**: $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$. It is an **E1** elimination, and the acid is a catalyst — consumed in step 1, regenerated in step 3.

Common slip · Two marks, three steps — so name the rate-determining step and show the **curly arrows**. Writing only the overall equation earns very little, because the question asked for the mechanism.

Carry out the following conversions in not more than 2 steps: **a)** But-1-ene to But-2-ene **b)** Benzene to Biphenyl

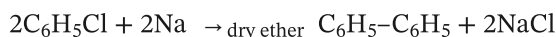
SOLUTION

How to see it · Both are two-step problems with the same shape: **put a handle on, then take it off again somewhere else**. In (a) the handle is a halogen that Markovnikov's rule places on the middle carbon; in (b) it is a halogen that lets two rings be joined by a metal.

- (a) But-1-ene → But-2-ene.** Adding HBr follows Markovnikov's rule, so Br goes to the *more substituted* carbon — C-2. Alcoholic KOH then eliminates, and Saytzeff's rule makes the *more substituted* alkene, which is but-2-ene.



- (b) Benzene → Biphenyl.** First chlorinate the ring, then couple two of the aryl halides with sodium in dry ether — the **Fittig reaction**.



Answer · (a) HBr, then alcoholic KOH. (b) $\text{Cl}_2/\text{AlCl}_3$, then Na in dry ether (Fittig).

Common slip · In (a), using **aqueous** KOH instead of alcoholic gives butan-2-ol by substitution, not the alkene. Alcoholic KOH eliminates, aqueous KOH substitutes — that one word decides the product.

Question 2 · Freezing point depression

2

Calculate the mass of ascorbic acid (Vitamin C, $\text{C}_6\text{H}_8\text{O}_6$) to be dissolved in 75 g of acetic acid to lower its freezing point by 1.5 °C. $K_f = 3.9 \text{ K kg mol}^{-1}$.

SOLUTION

How to see it · One formula, used backwards. $\Delta T_f = K_f \times \text{molality}$ tells you the molality; molality $\times$ kg of solvent gives moles; moles $\times$ molar mass gives the mass asked for.

$$\Delta T_f = K_f \cdot m \Rightarrow m = \frac{\Delta T_f}{K_f} = \frac{1.5}{3.9} = 0.3846 \text{ mol kg}^{-1}$$

- Solvent = 75 g = **0.075 kg** of acetic acid.
- Moles of ascorbic acid = $m \times \text{kg of solvent} = 0.3846 \times 0.075 = \mathbf{0.02885 \text{ mol}}$.
- Molar mass of $\text{C}_6\text{H}_8\text{O}_6 = (6 \times 12) + (8 \times 1) + (6 \times 16) = 72 + 8 + 96 = \mathbf{176 \text{ g mol}^{-1}}$.
- Mass = $0.02885 \times 176 = \mathbf{5.08 \text{ g}}$.

Answer · **5.08 g** of ascorbic acid.

A one-line version, worth memorising: $w_2 = (K_f \times w_1(\text{kg}) \times M_2) / \Delta T_f$. Note that ΔT_f is a difference, so 1.5 °C and 1.5 K are the same number — no conversion is needed.

B3 Chemical kinetics

[2]

A first-order reaction is 50% completed in 40 minutes at 300 K and in 20 minutes at 320 K. Calculate the activation energy of the reaction. [Given: $\log 2 = 0.3010$, $\log 4 = 0.6021$, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]

SOLUTION

How to see it · “50% completed” is the **half-life**. For a first-order reaction $t_{1/2} = 0.693/k$, so $k \propto 1/t_{1/2}$ — and that is all you need, because the Arrhenius equation only ever wants the **ratio** of the two rate constants. You never have to compute either k .

1. Halving the half-life doubles the rate constant:

$$\frac{k_2}{k_1} = \frac{t_{1/2(1)}}{t_{1/2(2)}} = \frac{40}{20} = 2$$

2. Now the two-temperature Arrhenius equation:

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \times \frac{T_2 - T_1}{T_1 T_2}$$

3. Rearrange for E_a and substitute $T_1 = 300 \text{ K}$, $T_2 = 320 \text{ K}$:

$$E_a = 2.303 \times 8.314 \times 0.3010 \times \frac{300 \times 320}{20}$$

$$E_a = 5.763 \times 4800 = 27664 \text{ J mol}^{-1}$$

Answer · $E_a = 27664 \text{ J mol}^{-1} \approx 27.66 \text{ kJ mol}^{-1}$

Common slip · The **$\log 4 = 0.6021$** in the data is a distractor. The ratio of rate constants here is 2, not 4 — the times halve, they do not quarter. Also: $T_1 T_2$ is a product (96000), not a sum.

Define order of reaction. How does order of a reaction differ from molecularity for a complex reaction?

SOLUTION

How to see it · The pair is the classic “experiment versus equation” distinction: **order is measured, molecularity is counted**. Every difference between them follows from that one sentence.

1. **Order of reaction.** The sum of the powers to which the concentration terms are raised in the *experimentally determined* rate law. For rate = $k[A]^x[B]^y$, the order is $x + y$.

	Order	Molecularity
Found by	experiment only	inspecting the balanced step
Value	may be 0, fractional or negative	a whole number, and always ≥ 1
Applies to	the overall reaction	one elementary step only
For a complex reaction	decided by the slowest step	has no meaning for the overall reaction

2. **The difference for a complex reaction.** A complex reaction happens in a sequence of elementary steps. Each *step* has a molecularity, but the overall reaction does not — you cannot add molecularities up. The overall **order**, however, is perfectly well defined: it is the order of the **rate-determining (slowest) step**, which is why it need not match the stoichiometric coefficients at all.

Answer · Order is experimental and can be zero, fractional or negative; molecularity is theoretical, a positive whole number, and is defined **only for an elementary step** — a complex reaction has an order but no molecularity.

*The standard example: decomposition of H_2O_2 written as $2H_2O_2 \rightarrow 2H_2O + O_2$ looks bimolecular, but is found to be **first** order, because the slow step involves only one H_2O_2 molecule.*

Question 4 · First-order timing

2

The rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to 1/10th of its initial value?

SOLUTION

How to see it · The integrated first-order law only ever needs the **ratio** $[A]_0/[A]$, never the concentrations themselves. Here that ratio is 10, and $\log 10 = 1$ — which is why this question has such a clean answer.

$$t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}$$

- $[A] = [A]_0/10$, so $[A]_0/[A] = 10$ and $\log 10 = 1$.
- $t = (2.303 / 60) \times 1 = 0.03838\text{ s}$.

Answer · $t = 3.84 \times 10^{-2}\text{ s}$ (about 38 milliseconds).

A rate constant of 60 s^{-1} is enormous — the half-life is $0.693/60 = 11.6\text{ ms}$ — so an answer in milliseconds is right, not a slip. The unit s^{-1} also confirms first order before you start.

After 24 hrs only 0.125 g out of the initial quantity of 1 g of a radioactive isotope remains behind. What is its half-life period?

SOLUTION

How to see it · Radioactive decay is first order, but do not reach for logarithms — look at the fraction first. 0.125 is exactly **1/8**, and $1/8 = (\frac{1}{2})^3$, so three half-lives have gone by. This is a counting problem.

1. Fraction left = $0.125 / 1 = 1/8$.
2. $1/8 = (\frac{1}{2})^3 \Rightarrow n = 3$ half-lives.
3. $1 \text{ g} \rightarrow 0.5 \text{ g} \rightarrow 0.25 \text{ g} \rightarrow 0.125 \text{ g}$ — three steps, confirming $n = 3$.
4. $t_{1/2} = \text{total time} / n = 24 / 3 = \mathbf{8 \text{ hours}}$.

Answer · $t_{1/2} = 8 \text{ hours}$.

The long way agrees: $k = (2.303/24) \log 8 = 0.0866 \text{ h}^{-1}$, and $t_{1/2} = 0.693/0.0866 = 8 \text{ h}$. Worth knowing, but the counting method is faster and safer under the clock.

Question 5 · Which is faster at S_N1 ?

2

In the following pairs of halogen compounds, which compound undergoes faster S_N1 reaction?

The two structures are drawn by hand on the scan. They are $(\text{CH}_3)_3\text{C}-\text{Cl}$ (2-chloro-2-methylpropane, tert-butyl chloride) and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{Cl}$ (1-chlorobutane, n-butyl chloride) — the two isomers of $\text{C}_4\text{H}_9\text{Cl}$ that this comparison is always set on.

SOLUTION

How to see it · S_N1 is two steps and the **first** one — the halide leaving to make a carbocation — is the slow one. So the rate depends on *nothing but the stability of the carbocation*. The nucleophile does not appear in the rate law at all.



1. $(\text{CH}_3)_3\text{C}-\text{Cl}$ ionises to the **tertiary** carbocation $(\text{CH}_3)_3\text{C}^+$. Three alkyl groups feed electron density in by **+I** and there are **nine α -hydrogens available for hyperconjugation**, so the charge is well spread out and the ion is comparatively stable.
2. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{Cl}$ would have to ionise to a **primary** carbocation, with one alkyl group and only two α -hydrogens. That is far too unstable to form — a primary halide reacts by S_N2 instead.
3. Carbocation stability runs $3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$, so the tertiary halide is faster — by a very wide margin, not a small one.

Answer · **tert-butyl chloride, $(\text{CH}_3)_3\text{C}-\text{Cl}$** , undergoes S_N1 much faster, because it forms a stable tertiary carbocation in the rate-determining step.

Common slip · The order is **exactly reversed for S_N2** , where the nucleophile must attack the carbon from behind: there the unhindered primary halide is fastest and the tertiary one barely reacts. Always check which mechanism the question names before quoting an order.

B6 Phenols / Ethers

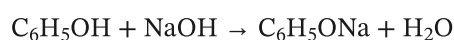
[2]

Write down the balanced chemical equation for the following name reaction: **a)** Kolbe's Reaction **b)** Williamson's ether synthesis.

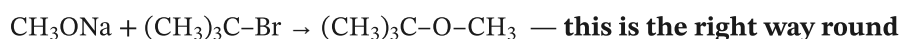
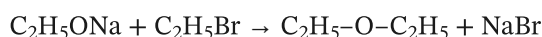
SOLUTION

How to see it · Both are one-line equations, but each has one condition that carries the mark. In Kolbe's it is that you must start from the **phenoxide**, not phenol; in Williamson's it is that the halide must be **primary**.

1. **(a) Kolbe's reaction.** Phenol is not nucleophilic enough to attack CO_2 , so it is first converted to sodium phenoxide with NaOH . The phenoxide then attacks CO_2 at the *ortho* position, and acidification gives salicylic acid.



2. **(b) Williamson's ether synthesis.** An alkoxide (or phenoxide) does $\text{S}_{\text{N}}2$ on an alkyl halide. Both symmetrical and unsymmetrical ethers can be made this way.



Answer · (a) sodium phenoxide + CO_2 at 400 K / 4–7 atm, then H^+ , gives salicylic acid. (b) $\text{R--ONa} + \text{R'--X} \rightarrow \text{R--O--R'} + \text{NaX}$, with R'--X primary.

Common slip · For an unsymmetrical ether, put the bulky group on the **alkoxide** and the small one on the **halide**. The reverse — a tertiary halide with an alkoxide — gives an alkene by elimination, because the alkoxide acts as a base. That is exactly the reaction Section C Q12(a) is about, run in the other direction.

B7 Alcohols

[2]

Name the reagent used in the following reactions: **a)** Benzyl alcohol to benzoic acid. **b)** Oxidation of primary alcohol to carboxylic acid.

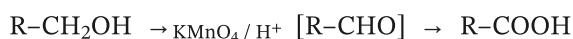
SOLUTION

How to see it · Both are the **full** oxidation of a primary alcohol — all the way to the acid, not stopping at the aldehyde — so both want a *strong* oxidising agent. This is the mirror image of item (iii) in Section A, where the answer had to be a mild one.

1. **(a) Benzyl alcohol → benzoic acid.** Reagent: **acidified KMnO_4** ($\text{KMnO}_4/\text{H}_2\text{SO}_4$), or alkaline KMnO_4 followed by H_3O^+ . Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ does equally well.



2. **(b) Primary alcohol → carboxylic acid.** The same reagent: **acidified KMnO_4** or acidified $\text{K}_2\text{Cr}_2\text{O}_7$.



Answer · Both: **acidified KMnO_4** (or acidified $\text{K}_2\text{Cr}_2\text{O}_7$). The aldehyde is formed first but is oxidised on without being isolated.

Common slip · Do **not** answer PCC here. PCC stops at the aldehyde, which is precisely why it was the right answer in Section A (iii) and is the wrong answer in this one. Read what the product is before choosing the oxidant.

Section C · 18 marks

Six questions of three marks. Four carry an internal choice; both alternatives are worked.

Question 8 · Proteins and amino acids

3

C8 Biomolecules

[3]

a) Differentiate between globular and fibrous protein. **b)** Give one example of essential and non-essential amino acids.

SOLUTION

How to see it · The two protein classes differ in **shape**, and everything else — solubility, function, whether heat destroys them — follows from shape. A ball of chain with its polar groups outside dissolves; a bundle of parallel threads held by hydrogen bonds does not. For (b), ‘essential’ means *the body cannot make it*, not *the body does not need it*.

	Globular protein	Fibrous protein
Shape	chain coiled into a compact, roughly spherical ball	chains lie parallel, held side by side in long threads
Held by	hydrogen bonds and other weak forces, easily disturbed	strong hydrogen bonds and disulphide bridges
Solubility in water	soluble	insoluble
Stability	denatured easily by heat, acid or alkali	tough and stable
Function	dynamic — enzymes, transport, hormones	structural — skin, hair, muscle
Examples	insulin, haemoglobin, albumin	keratin, myosin, collagen, fibroin

- (b) Essential amino acid** — one the human body *cannot synthesise*, so it must come from the diet. Example: **valine** (also leucine, isoleucine, lysine, phenylalanine, methionine, threonine, tryptophan, histidine, arginine).
- Non-essential amino acid** — one the body can make for itself. Example: **glycine** (also alanine, serine, aspartic acid, glutamic acid).

Answer · The table above; plus **valine** (essential) and **glycine** (non-essential).

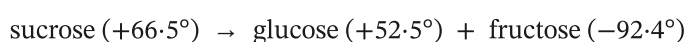
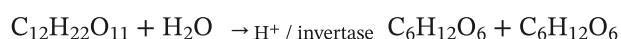
Ten of the twenty amino acids are essential for humans. All the amino acids in proteins are α -amino acids and, except for glycine, all are optically active — glycine is the exception because its side chain is another hydrogen, so its α -carbon is not a stereocentre.

Which sugar is called invert sugar? Why is it called so? Define peptide linkage?

SOLUTION

How to see it · ‘Invert’ is not about taste or structure — it is about the **sign of the optical rotation**, which literally inverts during the hydrolysis. Work it out by adding up the rotations of the products.

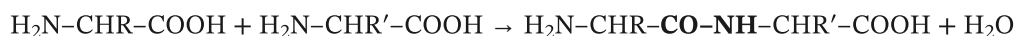
1. **Invert sugar** is the equimolar mixture of D-(+)-glucose and D-(−)-fructose obtained by hydrolysing **sucrose**.
2. Sucrose itself is **dextrorotatory**, $[\alpha]_D = +66.5^\circ$.
3. On hydrolysis it gives glucose, $+52.5^\circ$, and fructose, -92.4° . Fructose rotates left more strongly than glucose rotates right, so the mixture is net **laevorotatory**, about -19.9° .
4. The sign has gone from **+** to **−** — the rotation is *inverted*. Hence ‘invert sugar’, and the enzyme that does it is called **invertase**.



$$\text{mixture: } \frac{(+52.5^\circ) + (-92.4^\circ)}{2} = -19.9^\circ \text{ (laevorotatory)}$$

The products are formed in **equimolar** amounts, so the rotation of the mixture is the **mean** of the two specific rotations, not their sum. Adding them would give -39.9° , which is not what is observed.

5. **Peptide linkage**. The amide bond **−CO−NH−** formed between the **−COOH** group of one α -amino acid and the **−NH₂** group of the next, with the **loss of a molecule of water**.



Answer · **Sucrose** gives invert sugar; it is so called because hydrolysis inverts the rotation from $+66.5^\circ$ to -19.9° . A **peptide linkage** is the **−CO−NH−** amide bond between two amino acids, formed with loss of water.

The peptide C−N bond has partial double-bond character from resonance, so it cannot rotate freely and the whole **−CO−NH−** unit is planar. That rigidity is what makes the α -helix and β -pleated sheet possible.

Question 9 · Arrhenius from a graph

3

C9 Chemical kinetics

[3]

a) The slope of the line in the graph of $\log k$ vs $1/T$ for a reaction is -5841 K . Calculate the energy of activation. **b)** Define temperature coefficient.

SOLUTION

How to see it · Take logs of the Arrhenius equation and it becomes the equation of a straight line, $y = mx + c$, with $\log k$ on the y axis and $1/T$ on the x axis. Then just read the slope off and equate.

$$k = Ae^{-E_a/RT} \Rightarrow \log k = \log A - \frac{E_a}{2.303 R} \cdot \frac{1}{T}$$

1. Comparing with $y = mx + c$: slope $= -E_a / 2.303R$.
2. So $E_a = -2.303 \times R \times \text{slope} = -2.303 \times 8.314 \times (-5841)$.

$$E_a = 2.303 \times 8.314 \times 5841 = 111838 \text{ J mol}^{-1}$$

Answer · $E_a = 111838 \text{ J mol}^{-1} \approx 111.8 \text{ kJ mol}^{-1}$

3. **(b) Temperature coefficient.** The **ratio of the rate constants at two temperatures ten degrees apart** — conventionally 308 K and 298 K, or 318 K and 308 K.

$$\text{temperature coefficient} = \frac{k_{(T+10)}}{k_T} = \frac{k_{308}}{k_{298}}$$

Its value for most reactions is between **2 and 3** — that is the familiar statement that a reaction roughly doubles in rate for every 10 K rise. The Heritage 2016–17 paper sets exactly this idea in reverse, giving a coefficient of 1.75 and asking for E_a .

Common slip · The slope is **negative** and E_a is **positive** — the two minus signs cancel. Reporting a negative activation energy is the standard slip here. Also note the slope's unit is K, which is what makes E_a come out in J mol^{-1} .

C9 OR Chemical kinetics

[3]

The rate of a reaction depends upon the temperature and is quantitatively expressed as $k = Ae^{-E_a/RT}$. **a)** If a graph is plotted between $\log k$ vs $1/T$, write the expression for the slope. **b)** If $E_{a1} = 40 \text{ J/mol}$ and $E_{a2} = 80 \text{ J/mol}$, which of the two has a larger value of the rate constant?

SOLUTION

How to see it · Part (a) is the same linearisation as above. Part (b) needs only the sign of the exponent: E_a sits in $e^{-E_a/RT}$ with a **minus**, so a **bigger** barrier gives a **smaller** k . A high activation energy means a slow reaction.

1. **(a)** Taking $\log_{10}$ of both sides:

$$\log k = \log A - \frac{E_a}{2.303 R} \cdot \frac{1}{T} \Rightarrow \text{slope} = -\frac{E_a}{2.303 R}$$

2. The intercept on the $\log k$ axis is $\log A$.
3. **(b)** Both reactions are at the same temperature, and A is taken as comparable, so k is decided entirely by $e^{-E_a/RT}$.
4. $E_{a1} = 40 \text{ J/mol}$ is the **smaller** barrier, so $-E_a/RT$ is less negative and the exponential is larger.

Answer · **(a)** slope $= -E_a/2.303R$. **(b)** The reaction with $E_{a1} = 40 \text{ J mol}^{-1}$ has the larger rate constant — lower activation energy, faster reaction.

The energies quoted are in J mol^{-1} , not kJ — 40 J mol^{-1} is an absurdly low barrier for a real reaction. It does not affect the comparison, which is what is being asked.

C10 Haloalkanes / Phenols

[3]

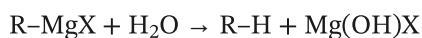
Give reason for the following: (any three)

(i) Alkyl halides, though polar, are immiscible with water. (ii) Racemic mixture is optically inactive. (iii) Grignard's reagents should be prepared under anhydrous conditions. (iv) Ethyl iodide undergoes S_N2 reaction faster than ethyl bromide. (v) Ortho-nitrophenol has lower boiling point than para-nitrophenol.

SOLUTION

How to see it · Only three are needed, but all five are worked here. Four of the five turn on **hydrogen bonding or bond strength**; (ii) is the odd one out and is about symmetry.

1. **(i) Alkyl halides are immiscible with water.** Dissolving something in water means breaking the extensive hydrogen-bond network water has with itself. An alkyl halide is polar, but it has **no hydrogen attached to N, O or F**, so it cannot form hydrogen bonds with water to pay that cost back. The new dipole–dipole interactions are far weaker than the $O-H\cdots O$ bonds destroyed, so mixing is not favourable.
2. **(ii) A racemic mixture is optically inactive.** It contains the two enantiomers in **exactly equal amounts**. They are mirror images, so they rotate plane-polarised light through equal angles in *opposite* directions. The rotations cancel exactly and the net rotation is zero — this is called **external compensation**. Note the mixture is still made of optically active molecules; it is the *mixture* that is inactive, unlike a meso compound where the cancellation is internal.
3. **(iii) Grignard reagents need anhydrous conditions.** The carbon of $R-MgX$ carries a substantial negative charge — it is effectively a carbanion, and therefore a very strong base. Water is acidic enough to protonate it instantly, destroying the reagent and giving the alkane:



4. Even moisture on the glassware is enough, which is why the ether and apparatus must be dried. Any compound with an active hydrogen — alcohols, amines, acids — does the same thing.
5. **(iv) Ethyl iodide is faster than ethyl bromide at S_N2 .** Both have the same carbon skeleton, so sterics are identical and the only difference is the leaving group. The **C–I bond is the weakest** of the carbon–halogen bonds (iodine is largest, so orbital overlap with carbon is poorest), and I^- is the most stable, most polarisable halide ion — the best leaving group. Reactivity runs **$R-I > R-Br > R-Cl > R-F$** .
6. **(v) o-Nitrophenol boils lower than p-nitrophenol.** In the *ortho* isomer the $-NO_2$ and $-OH$ groups are adjacent, so the molecule hydrogen bonds **to itself** — *intramolecular* hydrogen bonding, called chelation. That satisfies the $-OH$ internally, so the molecules do not associate with one another. In the *para* isomer the two groups are too far apart for that, so it forms **intermolecular** hydrogen bonds instead, associating into larger units. More energy is needed to separate those, so p-nitrophenol boils higher.

Answer · (i) no hydrogen bonding with water; (ii) equal and opposite rotations cancel by external compensation; (iii) water protonates the carbanion-like carbon and destroys the reagent; (iv) C–I is the weakest bond and I^- the best leaving group; (v) *ortho* has intramolecular H-bonding, *para* has intermolecular.

(v) has two useful consequences worth quoting: o-nitrophenol is **steam volatile** and is **less soluble in water** than the para isomer — both for the same reason, that its $-OH$ is tied up internally. That is how the two are separated.

C11 Solutions

[3]

2 g of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 K. Molal depression constant for benzene is $4.9 \text{ K kg mol}^{-1}$. What is the percentage association of acid if it forms dimer in solution?

SOLUTION

How to see it · Three moves, always in this order. Get the **observed** molar mass from the freezing-point data; compare it with the **true** molar mass to get i ; then put i into the association expression. The clue that association is happening is that i comes out **less than 1** — fewer particles than you put in.

1. **Step 1 — observed molar mass** from $\Delta T_f = K_f \cdot w_2 \times 1000 / (M_2 \times w_1)$:

$$M_{2(\text{obs})} = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1} = \frac{4.9 \times 2 \times 1000}{1.62 \times 25} = \frac{9800}{40.5} = 242 \text{ g mol}^{-1}$$

2. **Step 2 — van't Hoff factor.** The true molar mass of $\text{C}_6\text{H}_5\text{COOH}$ is $(7 \times 12) + (6 \times 1) + (2 \times 16) = 84 + 6 + 32 = 122 \text{ g mol}^{-1}$.

$$i = \frac{M_{2(\text{normal})}}{M_{2(\text{observed})}} = \frac{122}{242} = 0.504$$

3. $i < 1$ confirms association — the observed molar mass is almost exactly **twice** the true one, which is what dimerisation does.
4. **Step 3 — degree of association.** For $2\text{A} \rightarrow \text{A}_2$, starting with 1 mol and letting a fraction α associate: A left = $1 - \alpha$, dimer formed = $\alpha/2$, so total = $1 - \alpha/2$.

$$i = 1 - \frac{\alpha}{2} \Rightarrow \alpha = 2(1 - i) = 2(1 - 0.504) = 0.992$$

Answer · $\alpha = 0.992$, i.e. the percentage association is **99.2%** — almost all the benzoic acid is present as the dimer.

*Benzoic acid dimerises in benzene through **two** $\text{O-H}\cdots\text{O}=\text{C}$ hydrogen bonds, forming a closed eight-membered ring. Benzene is non-polar and cannot solvate the $-\text{COOH}$ group, so the acid molecules pair up with each other instead — which is why the answer is so close to 100%.*

Common slip · For **association**, $i = 1 - \alpha(1 - 1/n)$, which for a dimer ($n = 2$) is $1 - \alpha/2$. For **dissociation** it is $i = 1 + \alpha(n - 1)$. Using the dissociation formula here gives a negative α , which is the signal that you have picked the wrong one.

C12 Ethers / Alcohols

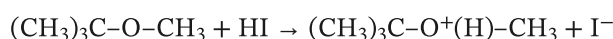
[3]

- (a) [Explain why] $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3$ on reaction with HI gives CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$ as the main products and not $(\text{CH}_3)_3\text{C}-\text{OH}$ and CH_3I .
- (b) Give a chemical test to distinguish between: i) Ethanol and tert. butyl alcohol. ii) Ethanol and methanol.

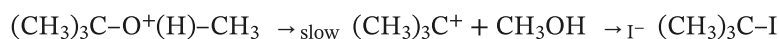
SOLUTION

How to see it · For (a), every ether cleavage starts the same way — the oxygen is protonated, so that the leaving group is a neutral alcohol rather than an alkoxide. The **question is which of the two C–O bonds then breaks**, and that is decided by whether a stable carbocation is available. Here one is.

1. **Step 1.** HI protonates the ether oxygen:



2. **Step 2.** The *tert*-butyl C–O bond breaks on its own, because it produces the **stable tertiary carbocation** $(\text{CH}_3)_3\text{C}^+$ — stabilised by +I from three methyl groups and by nine α -hydrogens through hyperconjugation. Methanol leaves as the neutral molecule. This is **S_N1**.
3. **Step 3.** I^- captures the carbocation, giving $(\text{CH}_3)_3\text{C}-\text{I}$.



4. **Why not the other way.** The alternative would need I^- to attack the methyl carbon by **S_N2**. That pathway is available in principle, but it is far slower than the spontaneous ionisation of a tertiary centre, so it does not compete — the **S_N1** route wins and the products are CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$.
5. **(b)(i) Ethanol vs *tert*-butyl alcohol — Lucas test.** Add Lucas reagent (conc. HCl + anhydrous ZnCl_2) at room temperature and watch for turbidity, which is the insoluble alkyl chloride separating out.
6. • ***tert*-butyl alcohol (3°):** turbidity **immediately** — it forms a tertiary carbocation at once.
• **Ethanol (1°):** **no turbidity** at room temperature; it needs heating.
7. **(b)(ii) Ethanol vs methanol — iodoform test.** Warm with I_2 and NaOH (or NaOI).
8. • **Ethanol** gives a **yellow precipitate** of iodoform, CHI_3 , with its characteristic smell.
• **Methanol** gives **no precipitate**.



Answer · (a) Protonation, then **S_N1** cleavage at the tertiary carbon because it gives a stable 3° carbocation. (b)(i) **Lucas test** — immediate turbidity with *tert*-butyl alcohol. (b)(ii) **Iodoform test** — yellow CHI_3 with ethanol only.

Common slip · The iodoform test needs the **$\text{CH}_3\text{CH}(\text{OH})-$** grouping (a methyl carbinol) or a methyl ketone. Methanol has no carbon next to the $-\text{OH}$ at all, so it cannot give iodoform — and neither can ethane-1,2-diol or propan-1-ol. This exact requirement decides Q13(a)(i) below.

C13 Alcohols / Phenols / Ethers

[3]

- (a) Give chemical tests to distinguish between: (i) Pentan-2-ol and Pentan-3-ol (ii) Methanol and Phenol
(b) Why are aromatic ethers o,p-directing?

SOLUTION

How to see it · Both tests in (a) work by finding a **structural feature present in one compound and absent in the other** — a methyl carbinol in the first pair, an aromatic –OH in the second. For (b), direction in electrophilic substitution is always decided by *where the electron density ends up*, which you find by drawing the resonance structures.

1. **(a)(i) Pentan-2-ol vs pentan-3-ol — iodoform test.** Write both structures out and look at the carbon bearing the –OH:
2. • **Pentan-2-ol**, $\text{CH}_3\text{--CH(OH)--CH}_2\text{CH}_2\text{CH}_3$, **has** the $\text{CH}_3\text{--CH(OH)--}$ grouping → warm with I_2/NaOH and a **yellow precipitate of CHI_3** forms.
• **Pentan-3-ol**, $\text{CH}_3\text{CH}_2\text{--CH(OH)--CH}_2\text{CH}_3$, has ethyl groups on both sides — no methyl carbinol → **no precipitate**.
3. **(a)(ii) Methanol vs phenol — neutral FeCl_3 .**
4. • **Phenol** gives an intense **violet colouration** (an iron–phenoxide complex).
• **Methanol** gives **no colour**.
5. An equally good alternative for (a)(ii): add **bromine water** — phenol gives a **white precipitate** of 2,4,6-tribromophenol; methanol does nothing. Or test with NaOH: phenol is acidic enough to dissolve as sodium phenoxide, methanol is not.
6. **(b) Why aromatic ethers are o,p-directing.** The oxygen of –OR carries lone pairs, and one of them is delocalised into the ring — a **+R (mesomeric) effect**. Drawing the resonance structures of anisole shows where that electron density goes:
7. The negative charge appears at the **ortho and para** positions and **never at meta**. Those two positions are therefore the electron-rich ones, so an incoming **electrophile** attacks there.
8. The same argument seen through the intermediate: attack at *ortho* or *para* gives an arenium ion that can be stabilised by the oxygen lone pair (an extra resonance structure with a complete octet on every atom), whereas attack at *meta* cannot be. The lower-energy intermediate forms faster.

–O–R → ring positions bearing δ^- : **2, 4 and 6** (*ortho, para*)

Answer · (a)(i) **Iodoform test** — pentan-2-ol gives yellow CHI_3 , pentan-3-ol does not. (a)(ii) **Neutral FeCl_3** — phenol gives a violet colour, methanol does not. (b) The –OR oxygen donates a lone pair into the ring by **+R**, raising the electron density at the *ortho* and *para* positions, so the electrophile goes there.

–OR is **activating** as well as directing: anisole brominates without any catalyst at all. The –I effect of oxygen opposes the +R effect but is much the weaker of the two, which is why the group activates rather than deactivates.

Convert: **i)** Aniline to phenol. **ii)** Ethyl magnesium chloride to propanol. **iii)** Phenol to picric acid.

SOLUTION

How to see it · Three standard conversions. (i) goes through the diazonium salt — the way almost every aniline conversion goes. (ii) is a Grignard addition, and the **carbon count decides which carbonyl compound to use**. (iii) needs care because concentrated HNO_3 also oxidises phenol.

1. **(i) Aniline** → **phenol**. Diazotise at 273–278 K, then warm the diazonium salt with water.



2. **(ii) Ethyl magnesium chloride** → **propanol**. Count the carbons: the Grignard brings **two** and propan-1-ol has **three**, so the carbonyl compound must supply exactly one carbon — that is **formaldehyde**. Formaldehyde with any Grignard always gives a *primary* alcohol.



3. **(iii) Phenol** → **picric acid** (2,4,6-trinitrophenol). The direct route is conc. HNO_3 /conc. H_2SO_4 , but phenol is easily oxidised by nitric acid and the yield is poor. The **preferred laboratory route sulphonates first**: the $-\text{SO}_3\text{H}$ groups protect the ring from oxidation and are then displaced by the nitro groups.



Answer · (i) NaNO_2/HCl at 273–278 K, then H_2O and warm. (ii) HCHO , then H_3O^+ . (iii) conc. H_2SO_4 , then conc. HNO_3 (or conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ directly).

*Picric acid is a genuinely strong acid — $\text{p}K_a \approx 0.4$, comparable to a mineral acid — because three nitro groups delocalise the phenoxide charge. It is the one phenol that dissolves in NaHCO_3 **with effervescence**, which ordinary phenols cannot do.*

Section D · 15 marks

Three questions of five marks. Q16 carries an internal choice; both alternatives are worked.

Question 14 · Identify A, B, C

5

D14 Alcohols / Haloalkanes

[5]

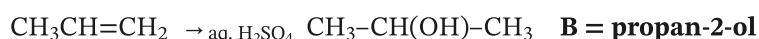
a) An organic compound 'A' having molecular formula C_3H_6 on treatment with aqueous H_2SO_4 gives 'B' which on treatment with conc $HCl/ZnCl_2$ gives 'C'. The compound on treatment with ethanolic KOH gives back the compound 'A'. Identify A, B and C.

b) How can propan-2-one be converted into tert-butyl alcohol?

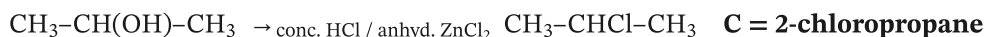
SOLUTION

How to see it · Start with the formula. C_3H_6 has one degree of unsaturation (C_nH_{2n}), so A is an alkene or a ring — and since it reacts with aqueous acid, it is the alkene **propene**. After that, every step is a named reaction you already know, and the last one is a **check**: getting A back proves the chain never changed length.

1. **A = propene**, $CH_3-CH=CH_2$. C_3H_6 , one double bond.
2. **A → B: acid-catalysed hydration**. Water adds across the double bond following **Markovnikov's rule** — the OH goes to the more substituted carbon (C-2), because that route goes through the more stable secondary carbocation.



3. **B → C: Lucas reagent**. Conc. HCl with anhydrous $ZnCl_2$ replaces the $-OH$ by $-Cl$. Propan-2-ol is a secondary alcohol, so it reacts in about five minutes.

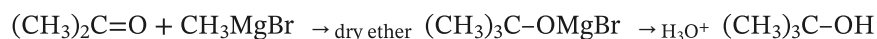


4. **C → A: dehydrohalogenation**. Alcoholic KOH eliminates HCl and regenerates the alkene — which is the confirmation the question builds in.



Answer · **A = propene** · **B = propan-2-ol** · **C = 2-chloropropane**

5. **b) Propan-2-one → tert-butyl alcohol**. Count the carbons first: propan-2-one has **three**, $(CH_3)_3C-OH$ has **four**. So exactly one carbon must be added — a **methyl** Grignard.
6. A ketone plus a Grignard reagent always gives a **tertiary** alcohol, which is what is wanted here.



Answer · **b)** React propan-2-one with **methyl magnesium bromide** in dry ether, then hydrolyse the adduct with dilute acid. Product: 2-methylpropan-2-ol (*tert*-butyl alcohol).

Common slip · The Grignard rule worth memorising: **HCHO** → **1° alcohol**, **any other aldehyde** → **2° alcohol**, **a ketone** → **3° alcohol**. And the hydrolysis step is not optional — without H_3O^+ you have only the magnesium alkoxide, which is not the alcohol.

D15 Chemical kinetics

[5]

(a) For $2A + B \rightarrow C + D$ the following results have been obtained during the kinetic studies of the reaction. Determine the rate law and the rate constant.

(b) The conversion of the molecule X to Y follows second order kinetics. If the concentration of X is increased to three times, how will it affect the rate of formation of Y?

SOLUTION

How to see it · The method of initial rates works by **changing one concentration at a time**. Find two experiments in which [B] is the same and only [A] differs — the ratio of rates then gives the order in A alone. Then do the same for B. Never try to use all four rows at once.

Exp	[A] / mol L ⁻¹	[B] / mol L ⁻¹	Initial rate / mol L ⁻¹ s ⁻¹
1	0.1	0.1	6.0×10^{-3}
2	0.3	0.2	7.2×10^{-2}
3	0.3	0.4	2.88×10^{-1}
4	0.4	0.1	2.40×10^{-2}

The paper prints experiment 2 as 7×10^{-2} . It must be 7.2×10^{-2} : with 7.2 all four experiments give the same $k = 6.00$, whereas 7.0 gives 5.83 for experiment 2 alone and no rate law fits all four rows.

- Let the rate law be $\text{rate} = k[A]^x[B]^y$.
- Order in A** — compare experiments **1 and 4**, where [B] is 0.1 in both:

$$\frac{\text{rate}_4}{\text{rate}_1} = \frac{2.40 \times 10^{-2}}{6.0 \times 10^{-3}} = 4 \quad \text{and} \quad \frac{[A]_4}{[A]_1} = \frac{0.4}{0.1} = 4$$

$$4^x = 4 \Rightarrow x = 1$$

- Order in B** — compare experiments **2 and 3**, where [A] is 0.3 in both:

$$\frac{\text{rate}_3}{\text{rate}_2} = \frac{2.88 \times 10^{-1}}{7.2 \times 10^{-2}} = 4 \quad \text{and} \quad \frac{[B]_3}{[B]_2} = \frac{0.4}{0.2} = 2$$

$$2^y = 4 \Rightarrow y = 2$$

Answer · Rate law: $\text{rate} = k[A][B]^2$ — first order in A, second order in B, **overall order 3**.

- Rate constant.** Rearranging, $k = \text{rate} / ([A][B]^2)$. Using experiment 1:

$$k = \frac{6.0 \times 10^{-3}}{0.1 \times (0.1)^2} = \frac{6.0 \times 10^{-3}}{1.0 \times 10^{-3}} = 6.0$$

- Check it against the other three rows — a rate law is only established if k is the same for every experiment:

Exp	[A][B] ²	rate	$k = \text{rate} / [A][B]^2$
1	1.0×10^{-3}	6.0×10^{-3}	6.00
2	1.2×10^{-2}	7.2×10^{-2}	6.00
3	4.8×10^{-2}	2.88×10^{-1}	6.00
4	4.0×10^{-3}	2.40×10^{-2}	6.00

Answer · $k = 6.0 \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$ (equivalently $6.0 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$).

The unit follows from the overall order $n = 3$: k has units $\text{mol}^{1-n} \text{ L}^{n-1} \text{ s}^{-1} = \text{mol}^{-2} \text{ L}^2 \text{ s}^{-1}$. Quoting k without its unit loses a mark.

6. (b) $X \rightarrow Y$, second order, $[X]$ tripled. Second order means $\text{rate} = k[X]^2$.

$$\frac{\text{rate}_{\text{new}}}{\text{rate}_{\text{old}}} = \frac{k(3[X])^2}{k[X]^2} = 3^2 = \mathbf{9}$$

Answer · (b) The rate of formation of Y becomes **9 times** its original value.

Common slip · Note that the stoichiometric coefficients — the 2 in front of A — play **no part** in finding the order. Order comes from experiment only. Writing $\text{rate} = k[A]^2[B]$ off the balanced equation is the mistake this question is built to catch: the data say first order in A, not second.

D16 Electrochemistry

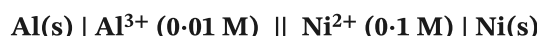
[5]

- a) Represent the cell in which the following reaction takes place: $2\text{Al(s)} + 3\text{Ni}^{2+}(0.1\text{ M}) \rightarrow 2\text{Al}^{3+}(0.01\text{ M}) + 3\text{Ni(s)}$. Calculate its emf if $E^\circ_{\text{cell}} = 1.41\text{ V}$.
- b) (i) State Kohlrausch law of independent migration of ions. (ii) State two advantages of the $\text{H}_2\text{-O}_2$ fuel cell over an ordinary cell.

SOLUTION

How to see it · Read the equation to see who is oxidised. Al goes from 0 to +3, so aluminium is **oxidised** and is therefore the **anode**; the anode is always written on the **left**. Then Nernst — and the only thing to be careful about is n and the direction of the reaction quotient.

1. **Cell representation.** Anode (oxidation, left) | its ion || cathode ion | cathode (reduction, right):



2. **Number of electrons.** $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$, doubled, gives 6 electrons; $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$, tripled, also 6. So $n = 6$.
3. **Nernst equation**, with Q built from the balanced reaction — products over reactants, solids omitted:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log \frac{[\text{Al}^{3+}]^2}{[\text{Ni}^{2+}]^3}$$

$$Q = \frac{(0.01)^2}{(0.1)^3} = \frac{1 \times 10^{-4}}{1 \times 10^{-3}} = 0.1 \Rightarrow \log Q = -1$$

$$E_{\text{cell}} = 1.41 - \frac{0.0591}{6} \times (-1) = 1.41 + 0.00985$$

Answer · $E_{\text{cell}} = 1.4199\text{ V} \approx 1.42\text{ V}$

The correction is tiny — about 10 mV — because $n = 6$ divides 0.0591 down and $\log Q$ is only -1 . It is positive, which makes sense: $Q < 1$ means there is less product than standard, so the cell is further from equilibrium and pushes harder than at standard conditions.

4. **b)(i) Kohlrausch's law of independent migration of ions.** At infinite dilution, when dissociation is complete, **each ion contributes a definite amount to the total molar conductivity of the electrolyte, irrespective of the other ion with which it is associated.** So the limiting molar conductivity of an electrolyte is the sum of the individual contributions of its cation and anion:

$$\Lambda^\circ_{\text{m}} = \nu_+ \lambda^\circ_+ + \nu_- \lambda^\circ_-$$

5. where ν_+ and ν_- are the numbers of cations and anions per formula unit. For example $\Lambda^\circ_{\text{m}}(\text{NaCl}) = \lambda^\circ(\text{Na}^+) + \lambda^\circ(\text{Cl}^-)$.
6. **b)(ii) Two advantages of the $\text{H}_2\text{-O}_2$ fuel cell** (any two):
- **Much higher efficiency** — about 70%, against roughly 40% for a thermal power plant, because the chemical energy is converted directly to electrical energy without going through heat.
 - **Pollution-free** — the only product is water, so there are no oxides of carbon, nitrogen or sulphur and no toxic heavy metals to dispose of.
 - **Continuous operation** — it never needs recharging and does not run down, as long as H_2 and O_2 are supplied.
 - The water produced is drinkable — which is why these cells were used in the Apollo spacecraft.

Answer · (a) $\text{Al} \mid \text{Al}^{3+}(0.01\text{ M}) \parallel \text{Ni}^{2+}(0.1\text{ M}) \mid \text{Ni}$, $E_{\text{cell}} = 1.42\text{ V}$. (b)(i) Λ°_{m} is the sum of the independent ionic contributions. (b)(ii) higher efficiency and no pollution (or: no recharging needed).

Common slip · The salt bridge is || (double line) and a phase boundary is | (single line). And n is 6 here, not 3 or 2 — it is the number of electrons in the **balanced overall** equation, which is why the two half-reactions had to be scaled to

match first.

D16 OR Electrochemistry

[5]

(a) The emf of a galvanic cell composed of two hydrogen electrodes is 0.16 volt at 25 °C. Calculate pH of the anode solution if the cathode is in a solution with pH = 1.

(b) Calculate the cell emf and ΔG° for the cell reaction at 25 °C for the cell: $\text{Zn(s)} \mid \text{Zn}^{2+}(0.0004 \text{ M}) \parallel \text{Cd}^{2+}(0.2 \text{ M}) \mid \text{Cd(s)}$. [$E^\circ: \text{Zn}^{2+}/\text{Zn} = -0.763 \text{ V}$; $\text{Cd}^{2+}/\text{Cd} = -0.403 \text{ V}$; $F = 96500 \text{ C mol}^{-1}$]

SOLUTION

How to see it · For (a): a **concentration cell**. Both electrodes are the same, so $E^\circ_{\text{cell}} = 0$ and the whole emf comes from the difference in H^+ concentration. The potential of a hydrogen electrode is $E = -0.0591 \text{ pH}$, so the cell emf is just $0.0591 \times (\text{difference in pH})$. For (b): standard Nernst, then $\Delta G^\circ = -nFE^\circ$.

1. (a) For a hydrogen electrode at 1 bar, $E = -0.0591 \text{ pH}$. So

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} = (-0.0591 \text{ pH}_c) - (-0.0591 \text{ pH}_a) = 0.0591 (\text{pH}_a - \text{pH}_c)$$

2. Substituting $E_{\text{cell}} = 0.16 \text{ V}$ and $\text{pH}_c = 1$:

$$0.16 = 0.0591 (\text{pH}_a - 1) \Rightarrow \text{pH}_a - 1 = \frac{0.16}{0.0591} = 2.707$$

Answer · pH of the anode solution = 3.71

The anode has the higher pH — the more dilute H^+ . That is the general rule for a concentration cell: oxidation happens in the dilute half, reduction in the concentrated one, because the cell runs so as to even the two out.

3. (b) **Standard cell potential**. Zn is the more negative, so it is oxidised — anode on the left, as the cell diagram already shows.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = (-0.403) - (-0.763) = +0.360 \text{ V}$$

4. The overall reaction is $\text{Zn(s)} + \text{Cd}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cd(s)}$, so $n = 2$. Nernst:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cd}^{2+}]}$$

$$Q = \frac{0.0004}{0.2} = 2 \times 10^{-3} \Rightarrow \log Q = -2.699$$

$$E_{\text{cell}} = 0.360 - 0.02955 \times (-2.699) = 0.360 + 0.0798$$

Answer · $E_{\text{cell}} = 0.4398 \text{ V} \approx 0.44 \text{ V}$

5. **Gibbs energy**. The standard value uses E° , not E :

$$\Delta G^\circ = -nFE^\circ_{\text{cell}} = -2 \times 96500 \times 0.360 = -69480 \text{ J mol}^{-1}$$

Answer · $\Delta G^\circ = -69480 \text{ J mol}^{-1} = -69.48 \text{ kJ mol}^{-1}$

If the question is read as wanting ΔG under the stated non-standard concentrations, use E_{cell} instead: $\Delta G = -2 \times 96500 \times 0.4398 = -84.87 \text{ kJ mol}^{-1}$. The degree sign in ΔG° is what selects E° , so quote the -69.48 kJ figure as the answer and add this as a note.

Common slip · ΔG° is negative here, which is what a **spontaneous** cell reaction must give — a positive E° always means a negative ΔG° . Also, F must be in coulombs and E in volts for the answer to come out in joules, not kilojoules.